# Jaypee Institute of Information Technology, Noida

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING AND INFORMATION TECHNOLOGY

Project Title: CampusVista - Smart Campus Navigation App

## Project Title: <u>Mock Interview System</u>


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Fundamentals of Mobile Application Development Lab

**Course Name:** Fundamentals of Data Ar [Course Code]

**Course Code:** 25B16CS211
**Program:** B. Tech. CSE
2nd Year 4th Semester
**2025 – 2026**

## 1. Project Title

CampusVista - Smart Campus Navigation App

## 2. Objective

- To design an Android application that helps users navigate a campus using a 2D map, checkpoints, searchable locations, and route guidance.
- To use Python FastAPI as the primary backend for search, shortest path route calculation, checkpoint processing, and crowd warning services.
- To provide a simple mobile navigation experience with map display, route preview, outdoor panorama images, and fallback support.

## 3. Requirements

**Software:** Android Studio, Java, XML, Python, FastAPI, SQLite, Android Emulator or Physical Android Device, Git/GitHub.

**Hardware:** Android smartphone or emulator, Laptop/PC for running Android Studio and Python backend, Camera-enabled device if recognition/camera feature is demonstrated.

## 4. Project Description

CampusVista is a smart campus navigation Android application. It helps users navigate inside a campus using a marked 2D campus map, searchable locations, checkpoints, generated routes, outdoor panorama images, and backend-supported routing/search logic.

The Android app is developed in Android Studio using Java and XML. It provides the mobile interface for setting current location, searching destinations, viewing the map, previewing routes, starting navigation, and viewing panorama images.

The Python FastAPI backend acts as the primary intelligence layer. It handles search, route calculation, checkpoint lookup, crowd warning logic, and data services. SQLite seed database stores checkpoints, edges, places, outdoor panos, crowd rules, recognition references, and search aliases.

## 5. Features

List of key features:

- Campus map displays marked checkpoints.
- User can select or set current location.
- User can search campus places such as gates, buildings, canteen, library, and outdoor areas.
- Shortest path route is generated using backend-supported logic.
- Crowd rules show warning popups/cards, but do not change the route.
- Outdoor panorama images are linked with checkpoints.
- Pano viewer supports checkpoint-to-checkpoint switching like simple Street View.
- App has fallback behavior if Python backend is unavailable.

## 6. Implementation Steps

1. Set up Android project in Android Studio.
2. Create UI screens using XML and Java activities.
3. Prepare campus data in Excel/SQLite.
4. Generate SQLite seed database.
5. Build Python FastAPI backend for search and routing.
6. Connect Android app to backend using API calls.
7. Display campus map, checkpoints, route, and pano images.
8. Test app on emulator/physical device.

## 7. Output

- Home screen with campus map.
- Search result screen.
- Route preview/navigation screen.
- Outdoor panorama viewer.
- Crowd warning popup if applicable.

## 8. Submission

- Source Code (zipped Android/Python project).
- Screenshots of working app.
- Short report of 2-3 pages containing:
  1. Project Title & Objective
  2. Description & Features
  3. Implementation Steps
  4. Screenshots of Output